

Questions with Answer Keys

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Q1: 24 Feb (Shift 2) - Single Correct

Which of the following equations represents a travelling wave?

(1) $y = Ae^{-x^2}(vt + \theta)$

(2) $y = A \sin(15x - 2t)$

(3) $y = Ae^x \cos(\omega t - \theta)$

(4) $y = A \sin x \cos \omega t$

Q2: 24 Feb (Shift 2) - Numerical

Two cars are approaching each other at an equal speed of 7.2 km/hr. When they see each other, both blow horns having frequency of 676 Hz. The beat frequency heard by each driver will be ____ Hz. [Velocity of sound in air is 340 m/s.]

Q3: 25 Feb (Shift 1) - Single Correct

A student is performing the experiment of resonance column. The diameter of the column tube is 6 cm. The frequency of the tuning fork is 504 Hz. Speed of the sound at the given temperature is 336 m/s. The zero of the metre scale coincides with the top end of the resonance column tube. The reading of the water level in the column when the first resonance occurs is :

(1) 13 cm

(2) 14.8 cm

(3) 16.6 cm

(4) 18.4 cm

Q4: 26 Feb (Shift 1) - Numerical

The mass per unit length of a uniform wire is 0.135 g/cm. A transverse wave of the form

$y = -0.21 \sin(x + 30t)$ is produced in it, where x is in meter and t is in second. Then, the expected value of tension in the wire is $x \times 10^{-2}$ N. Value of x is _____. (Round-off to the nearest integer)

Q5: 26 Feb (Shift 2) - Single Correct

A tuning fork A of unknown frequency produces 5beats/s with a fork of known frequency 340 HZ. When fork A filed, the beat frequency decreases to 2beats/s. What is the frequency of fork A?

(1) 342 Hz

(2) 335 Hz

(3) 338 Hz

(4) 345 Hz

Answer Key**Q1 (2)****Q2 (8)****Q3 (2)****Q4 (1215)****Q5 (2)**

Q1 (2)

$$Y = F(x, t)$$

For travelling wave y should be linear function of x and t and they must exist as $(x \pm vt)$

$$Y = A \sin(15x - 2t) \rightarrow \text{linear function in } x \text{ and } t$$

Q2 (8)



$$\text{Speed} = 7.2 \text{ km/h} = 2 \text{ m/s}$$

$$\text{Frequency as heard by A } f'_A = f_B \left(\frac{V+V_0}{V-V_s} \right)$$

$$f'_A = 676 \left(\frac{340+2}{340-2} \right)$$

$$f'_A = 684 \text{ Hz}$$

$$\therefore f_{\text{Beat}} = f'_A - f_B$$

$$= 684 - 676$$

$$= 8 \text{ Hz}$$

Q3 (2)

$$\lambda = \frac{V}{f} = \frac{336}{504} = 66.66 \text{ cm}$$

$$\frac{\lambda}{4} = 1 + e = 1 + 0.3 \text{ d}$$

$$= 1 + 1.8$$

$$16.66 = 1 + 1.8 \text{ cm}$$

$$I = 14.86 \text{ cm}$$

Q4 (1215)

$$y = -0.21 \sin(x + 30t)$$

$$v = \frac{\omega}{K} = \frac{30}{1} = 30 \text{ m/s}$$

$$v = \sqrt{\frac{T}{\mu}}$$

Hints and Solutions

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$$T = v^2 \times \mu$$

$$T = (30)^2 \times 0.135 \times 10^{-1} \quad \mu = 0.135 \text{ gm/cm}$$

$$T = 900 \times 0.135 \times 10^{-1} \quad \mu = 0.135 \times \frac{10^{-3} \text{ kg}}{10^{-2} \text{ m}} \frac{\text{kg}}{\text{m}}$$

$$T = 12.15 \text{ N}$$

$$T = 1215 \times 10^{-2} \text{ N}$$

$$X = 1215$$

Q5 (2)

Given

Before Filed :



So answer should be 335 Hz or 345 Hz.

After Filed :





After filed beat/sec decreases only in case of 335 Hz.